the massive coordinate sets that need to be stored for each detail. This level of nitty-gritty calculations allows a designer to begin a model as a cylinder and tweak it till it takes on the shape of a jet engine, with all its hollows, vents, passages, fans and other details.

As we can imagine and even see for ourselves, the ability to transfer these designs for collaborative work is invaluable. Unlike handmade manual designs, these designs can be easily augmented just as word processing softwares made it easier to edit text without redrafting, similarly CAD softwares make it possible for the effort placed in 3D design to be lesser and ultimately more efficient. All these features make 3D designs easy to share, customise and fabricate by the 3D printing community.

## Evolution of Commercial 3D Printing Software

The step towards an effective 3D printing software requires two elements - modelling design and surface design. As CAD programs evolved the major aspect was rooted in ensuring a reliant and accurate object modelling program. The need for surface design was never a major concern as that is a 3D printing issue and was never envisioned as the original purpose of CAD programs.

Only recently have CAD programs started including 3D printer requirements in their workflow. Since a traditional design file is limited in terms of the information it can convey to a printer, 3D printer based CAD programs need to include otherwise ignored information. The complexity that 3D printers are capable of include replicating not only physical objects made of manufacturing materials but also complex biological elements such as in the case of printing a human organ.

Since 3D printed objects are complete pieces rather than individual elements that need to be assembled later, the software needs to have the ability to not only map the surface of the object accurately but also interpret what lies below the surface of the virtual object such as hollows and patterns. To fulfil this purpose, there are two main categories of design software's that are used for 3D printing.

Solid modelling design software are used in the industrial and commercial segment by engineers and scientists. They employ the use of template library of pre-designed shapes such as spheres, cubes, cylinders etc which are then augmented, combined, cut and joint to create vastly complex solid models. The commercial CAD software's allow the ease and flexibility of using these shapes in congruence with their additive and subtractive design